Information and Route Choice: Analysis So Far

Part 1 (homogeneous, symmetric information) with a first look at Part 2 (asymmetric information). Results from the *simulation/route-counting* working tree, 23 October 2026.

1. Key findings

- **A little information beats both none and full information.** In all four networks the lowest cost occurs at a small α (0.01–0.16). At the best α, Pigou, Braess and Grid are within 0.3–4% of the system optimum. Both α = 0 and α = 1 are much worse.
- **Beyond that point, more information makes things worse.** Cost rises with α because agents' decisions become correlated: everyone sees the same crowd and makes the same choice. At α = 1, 100% of agents end on a single route in every network.
- **The quantity that matters is α × N**, the expected number of other agents each agent observes, rather than α alone. The best results occur when agents observe about 1–3 others. Plotted against α × N, the curves for N = 20, 50 and 80 line up near the minimum (closely for Pigou and Diamond, more loosely for Braess and Grid), so the "best α" shrinks as N grows while the best cost barely changes.
- **How the network is calibrated changes the conclusion.** In the earlier scenarios, where the crowded route sat exactly at break-even, cost was flat for every 0 < α < 1. In the current scenarios, where the crowd sits well above break-even, cost rises steadily with α. Both are correct for their calibration, so the write-up should show both.
- **Some results come from features of the model, not from agent behaviour.** These are tie-breaking at α = 1, information that only covers other agents' next step, and a single simultaneous round of reassessment. Each one measurably affects the results (Section 7).
- **Asymmetric information produces a real trend.** As the share of informed agents rises, cost falls steadily, and uninformed agents free-ride on informed ones. This preview used in-memory patches on the earlier break-even scenarios (Section 6).

2. Model as simulated

Every agent starts on its free-flow shortest path. Each timestep, every agent still travelling states its intended next edge. All agents then reassess at the same time (revision_percentage = 100). A reassessing agent sees each other agent's intention independently with probability α, estimates the traffic on each edge as *observed ÷ α*, adds itself to the edge it is considering, and picks the cheapest path. It only changes its plan if the new path is *strictly* cheaper, and exact ties are broken by networkx's path order. An edge costs $a + b \cdot x$, where x is the number of agents crossing it *in the same timestep*. Two routes therefore only congest each other on an edge they reach at the same hop.

Two consequences matter throughout. First, information only describes other agents' *next* step, so the rest of every path looks empty. Second, because the estimate is unbiased (divided by α), the decision boils down to "did I see more agents than I would expect to?"

3. Experimental setup

Network	Routes from origin to destination	Base N	Variants	Notes
Pigou	2 (1-2-4 congestible, 1-3-4 fixed cost 1)	80	N = 20, 50	Crowd costs 1.6 if everyone takes the top route; break-even is 1.0
Braess	3 (1-2-4, 1-3-4, 1-2-3-4)	80	N = 20, 50	Per-timestep congestion: 1-2-3-4 and 1-3-4 do not share edge 3→4
Diamond	4 parallel two-hop routes via nodes 2–5	80	N = 20, 50	Destination fixed to node 6 (it was the nonexistent node 9)

Network	Routes from origin to destination	Base N	Variants	Notes
Grid	6 monotone four-hop routes, 3×3 grid	80	N = 20, 50	N = 50 is the original grid network; N = 80 added for consistency

Every sweep uses 50 trials (seeds 3821–3870), with α from 0 to 1 in steps of 0.01 and every agent given the same α. For the N variants, every edge's congestion slope b is scaled so that $b \times N$ matches the base network. That keeps cost per agent comparable, so only N changes. The optimum costs below were computed under the simulator's per-timestep cost model, by exhaustive search (Pigou, Braess) or a convex integer search (Diamond, Grid). The two methods agree exactly on Pigou and Braess.

4. Results: symmetric information

Network (N = 80)	Cost per agent at α = 0	Best α (α × N)	Cost at best α	Optimum	Gap to optimum	Cost at α = 1
Pigou	1.600	0.03 (2.4)	0.847	0.844	0.3%	1.000
Braess	3.200	0.01 (0.8)	1.666	1.644	1.3%	2.600
Diamond	4.080	0.01 (0.8)	3.121	2.739	13.9%	4.120
Grid	6.110	0.03 (2.4)	5.228	5.131	1.9%	6.110

Table 1. Mean cost per agent at N = 80 (50 trials per point). Results for every N are in the Appendix.

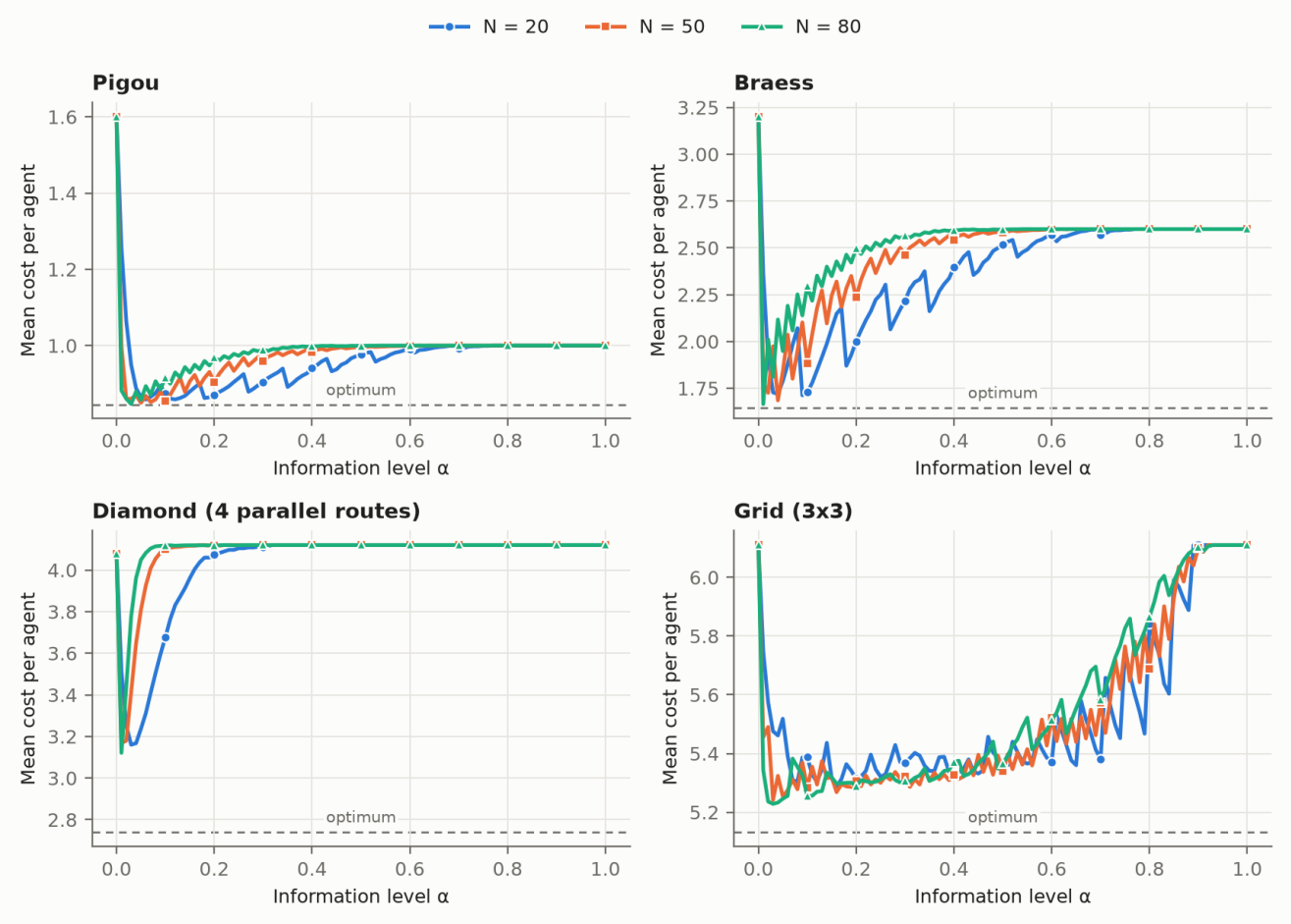


Figure 1. Mean cost per agent against information level α, for N = 20, 50 and 80. The dashed line is the system optimum. Every network has a sharp minimum at small α, then gets worse as α rises.

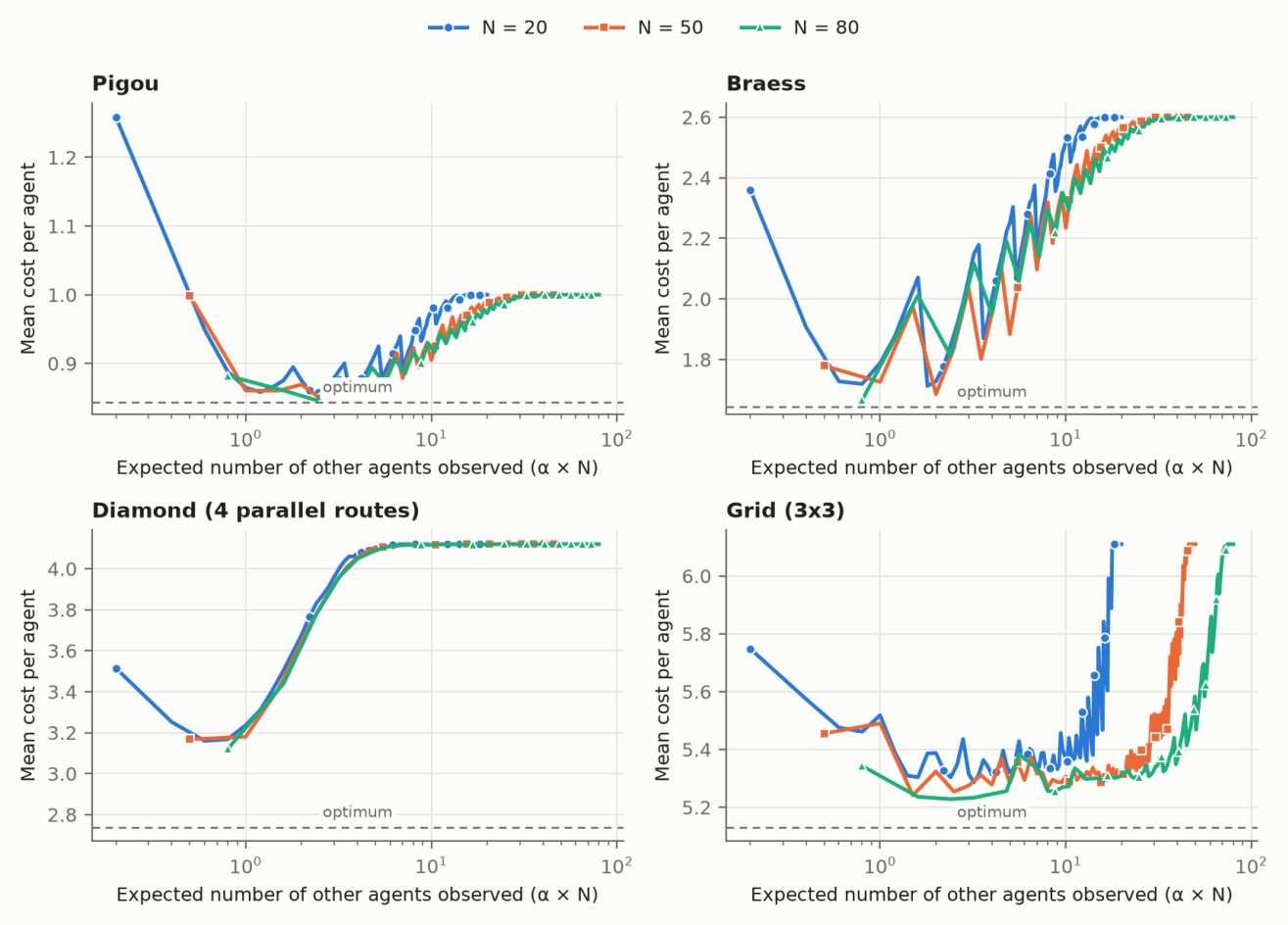


Figure 2. The same data plotted against α × N, the expected number of other agents observed, on a log scale. Around the minimum (α × N ≈ 1–3), Pigou and Diamond line up within about 0.05 per agent across N. Braess and Grid line up more loosely (differences up to about 0.2), partly because of sawtooth noise at N = 20. At high α × N, Grid's curves separate, because its rise depends on α itself: it starts once nearly everyone sees nearly everyone.

Network by network:

- **Pigou** gets within 0.3% of the optimum at α ≈ 0.03 (N = 80). Above α ≈ 0.5 every agent switches to the bottom route together, so cost settles at 1.0 per agent.
- **Braess** behaves the same way: best at α ≈ 0.01–0.09, then rises to 2.6 per agent as every agent herds onto 1-3-4.
- **Diamond** never gets closer than 14–15% to the optimum. Only two of its four routes are ever used: agents that switch all see the same crowd on 1-2-6 and nobody elsewhere, so every one of them picks the same next-best route (1-3-6). The optimum spreads agents over all four routes (23/21/19/17 at N = 80).
- **Grid** has a broad, flat minimum (α ≈ 0.03–0.5), then returns to the no-information cost at α ≥ 0.9. There, the switch rate drops to zero. Because information only covers the next step, a crowded first edge still looks cheaper than any detour, so nobody moves.

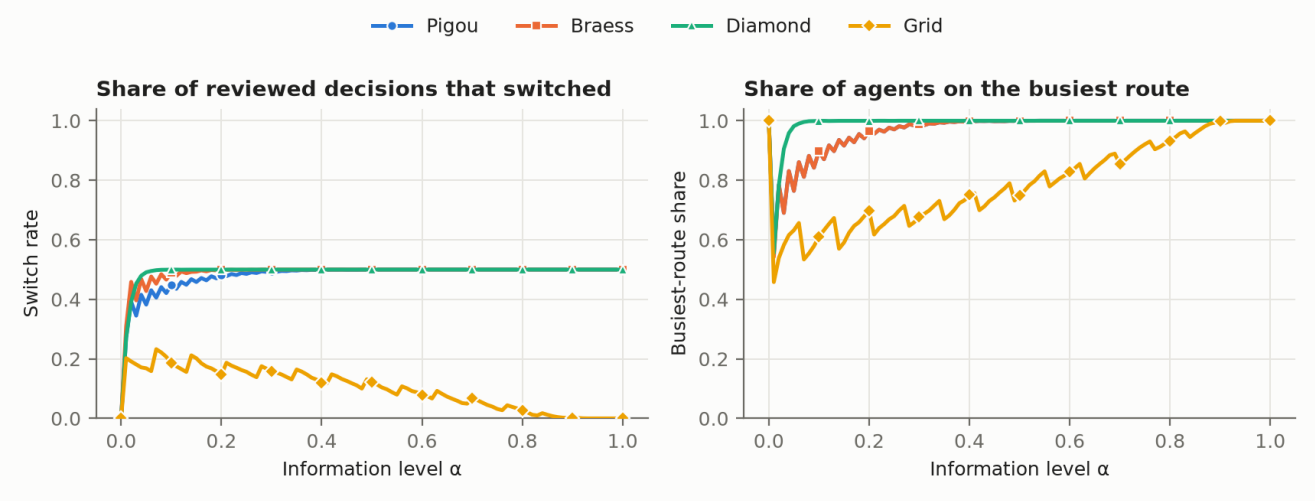


Figure 3. How agents behave at N = 80. Left: the share of reassessed decisions that changed. Pigou, Braess and Diamond level off at 0.5 because every agent makes two decisions (one per hop) and switches at the first one; the Pigou line is mostly hidden under Braess and Diamond. Right: the share of agents on the busiest route. It reaches 1.0 (everyone on one route) at moderate α for Pigou, Braess and Diamond, and approaches 1.0 at high α for Grid.

5. Why it happens

In Pigou an agent switches to the bottom route if $b \cdot (k/\alpha + 1) > 1$, where k is the number of agents it observes, drawn from Binomial(N − 1, α). With $b = 1.6/N$, the switching threshold works out to about $0.625 \cdot \alpha \cdot N$, while the expected count is about $\alpha \cdot N$. Every agent therefore tends to see more traffic than the threshold, and the more information it has, the more certain it is to switch. When α·N is small the counts are noisy, so agents switch independently and only a fraction of them move. As α·N grows, the chance of switching heads to 1 and the whole crowd moves together.

This model predicts Pigou's best point exactly. The optimum needs 55 of 80 agents on the bottom route, a 69% switch probability, which the binomial gives at α ≈ 0.03. The sweep's best point is α = 0.03. More generally, the predicted share of agents on the bottom route matches the measured share to within 0.015 at every α checked (0.01–0.3), e.g. 0.690 predicted vs 0.691 measured at α = 0.03.

The general mechanism. What decides the outcome is how *correlated* agents' decisions are, not how accurate any single agent's information is. Independent noise spreads agents across routes, and shared information moves them together. More information makes decisions more correlated, so it only helps until decisions stop being independent.

Earlier calibration: the crowd exactly at break-even

The earlier Pigou and Braess scenarios (top edge cost x/N) put the crowd exactly at break-even: everyone on the top route costs 1.0, the same as the bottom route. The switching threshold then equals the expected count, so each decision is close to a coin flip at any α, and a coin flip happens to give the optimal 50/50 split. The cost curve is flat for every $0 < \alpha < 1$. At α = 1 every agent sees exactly the break-even crowd, the strict > keeps everyone where they are, and cost returns to the selfish equilibrium (Figure 4).

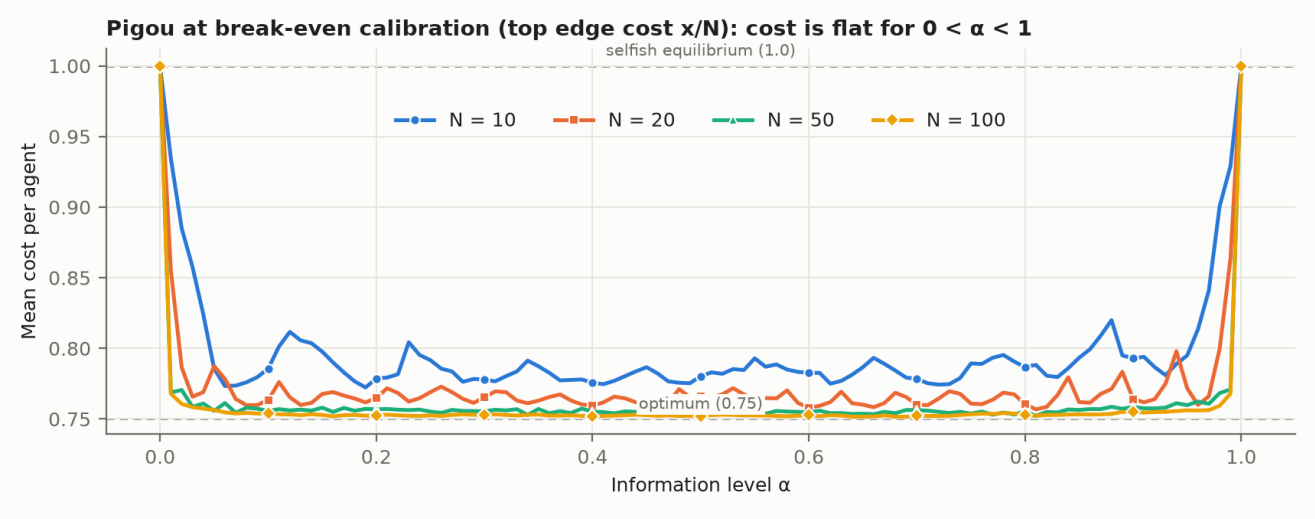


Figure 4. Earlier break-even calibration of Pigou (30 trials per point). The sawtooth comes from k being a whole number: the switching threshold jumps every 1/(N − 1) in α. The teeth shrink about 13 times from N = 10 to N = 100 and disappear at N = 100.

6. Robustness checks and an asymmetric preview

The checks below were run with in-memory patches (none of this code is in the repository) on the **earlier break-even calibration**. They show which results depend on the information model.

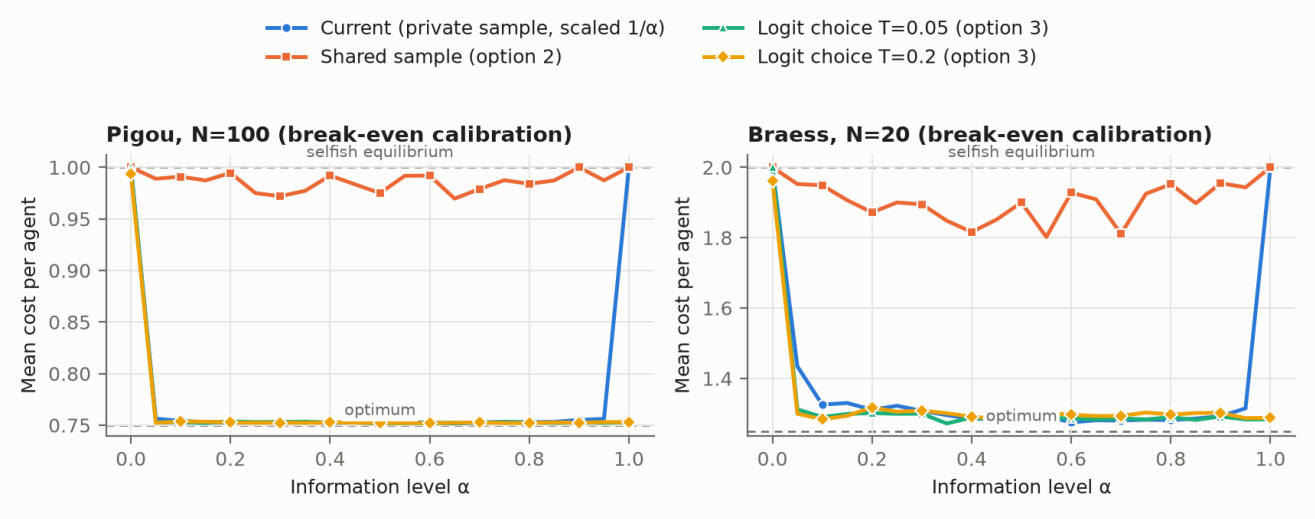


Figure 5. Alternative information models. **Shared sample:** every reassessing agent sees the same random sample, so they all decide the same way, and cost stays near the selfish equilibrium at every α. **Logit choice:** agents choose randomly with a preference for cheaper routes, and cost is near the optimum at every α > 0, including α = 1. Randomness in the choice does the same job as noise in the information. A third variant, which drops the ÷α scaling, produced no switching at all in these networks.

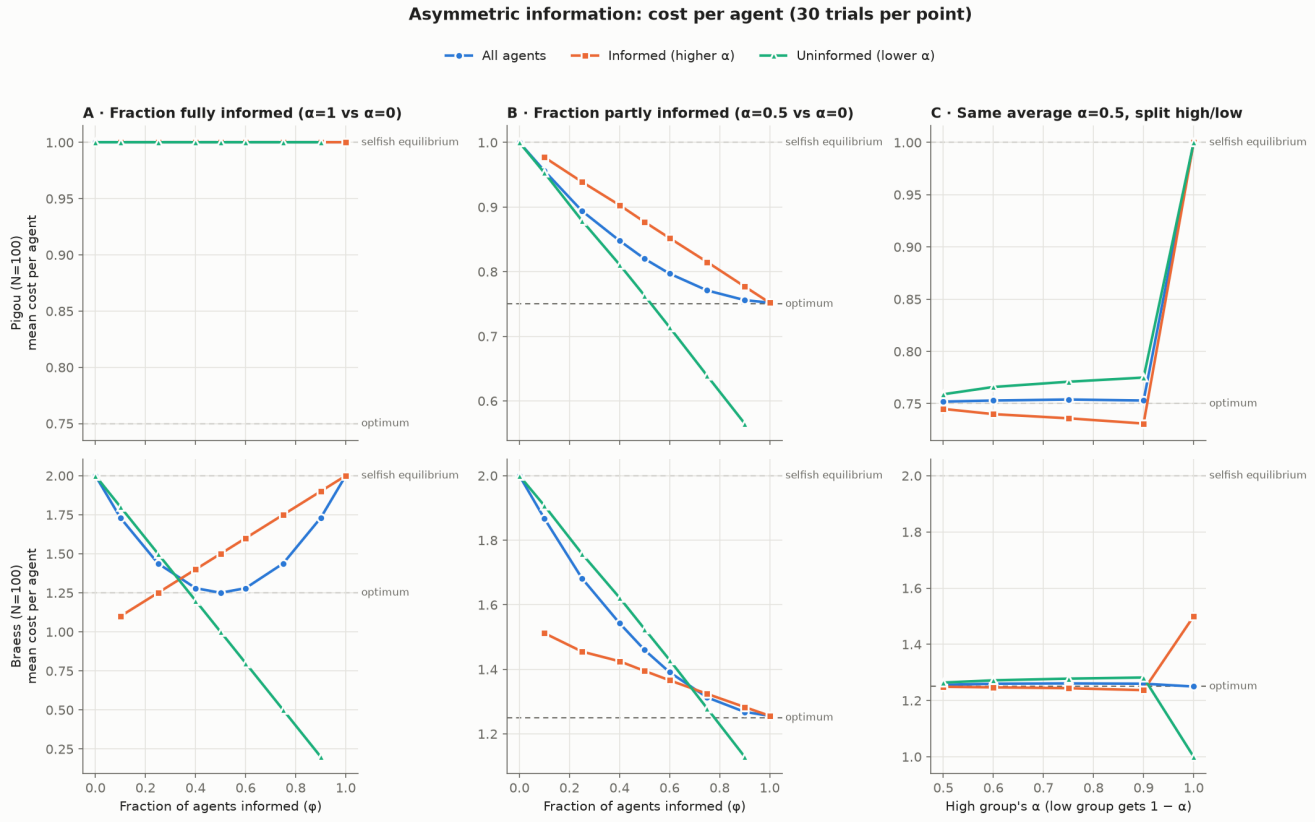


Figure 6. Asymmetric information on the break-even N = 100 scenarios (30 trials per point). **B (middle):** giving a growing fraction of agents α = 0.5, with the rest at α = 0, lowers cost steadily, and uninformed agents pay less than informed ones. **A (left):** with fully informed (α = 1) versus uninformed agents, Braess is a clean U-shape with the optimum at 50% informed, and past 25% the informed agents pay more. Pigou does not move at all, which is a tie-breaking effect. **C (right):** at a fixed average α = 0.5, how information is split between groups hardly matters, except at the (1, 0) split, which is again decided by tie-breaking.

7. Caveats and model issues

- **Tie-breaking.** Results at α = 1 in the break-even scenarios, and every asymmetric case with an α = 1 group, are decided by the strict > and networkx's path order. That order is arbitrary and not something the model intends.
- **Per-timestep congestion.** Routes of different lengths reach a shared edge at different timesteps, so they do not congest each other there. Braess here is therefore a timing-based variant, not the textbook paradox. Optima are computed under this rule.
- **Information only covers the next step.** Agents cannot see congestion further along a path. This produces Grid's "nobody switches" result at high α.
- **One round of reassessment, all at once.** Everyone reacts to the same starting intentions. This limits Diamond to two routes, and it is why shared or full information causes herding.
- **Data housekeeping.** Diamond's destination was fixed. The previous Grid results could not be reproduced from any committed code and have been replaced. The stale *results/grid/results.json* and *history.json* are left over from an older grid and should be removed.
- **The "best α" the sweep prints is noisy at N = 20**, because of the sawtooth. Report results against α × N, with N ≥ 50.

8. Recommended next steps

- Make random tie-breaking the default (or offer logit choice) before running Part 2, so α = 1 results reflect behaviour rather than path ordering.
- Run Part 2 using the multi-group agent generation on *simulation/asymmetric*, varying the fraction of informed agents, with the current (above break-even) calibration, and record cost for each group.

- Try staggered reassessment (revision_percentage < 100, or several rounds) to test whether accurate information helps once agents are not all reacting at the same moment.
- Move the in-memory experiments (alternative information models, asymmetric preview, N-scaling) into scripts in the repository so Sections 5–6 can be reproduced.

Appendix A. Results for every network and N

Scenario	N	α = 0	Best α	α × N	Cost at best α	Optimum	Gap	α = 1
pigou_N20	20	1.600	0.06	1.2	0.858	0.844	1.7%	1.000
pigou_N50	50	1.600	0.05	2.5	0.850	0.844	0.7%	1.000
pigou	80	1.600	0.03	2.4	0.847	0.844	0.3%	1.000
braess_N20	20	3.200	0.09	1.8	1.712	1.644	4.2%	2.600
braess_N50	50	3.200	0.04	2.0	1.685	1.644	2.5%	2.600
braess	80	3.200	0.01	0.8	1.666	1.644	1.3%	2.600
diamond_N20	20	4.080	0.03	0.6	3.160	2.740	15.3%	4.120
diamond_N50	50	4.080	0.01	0.5	3.168	2.739	15.6%	4.120
diamond	80	4.080	0.01	0.8	3.121	2.739	13.9%	4.120
grid_N20	20	6.110	0.16	3.2	5.287	5.132	3.0%	6.110
grid_N50	50	6.110	0.03	1.5	5.241	5.132	2.1%	6.110
grid	80	6.110	0.03	2.4	5.228	5.131	1.9%	6.110

Table A1. Mean cost per agent, 50 trials per point. The folder named after the network (e.g. *pigou*) is N = 80.

Appendix B. Reproducing the sweeps

From the repository root, with the project venv:

```
python -m src.simulation.symmetric_sweep --scenario scenarios/<name> --output results/<name> --trials 50
--alpha-step 0.01
```

Each results folder contains raw_results.csv (one row per run), summary.csv (one row per α), route_results.csv, edge_results.csv, alpha_sweep.png and alpha_routes.png. Figures 4–6 come from scratch experiments that are not in the repository yet (see Section 8).